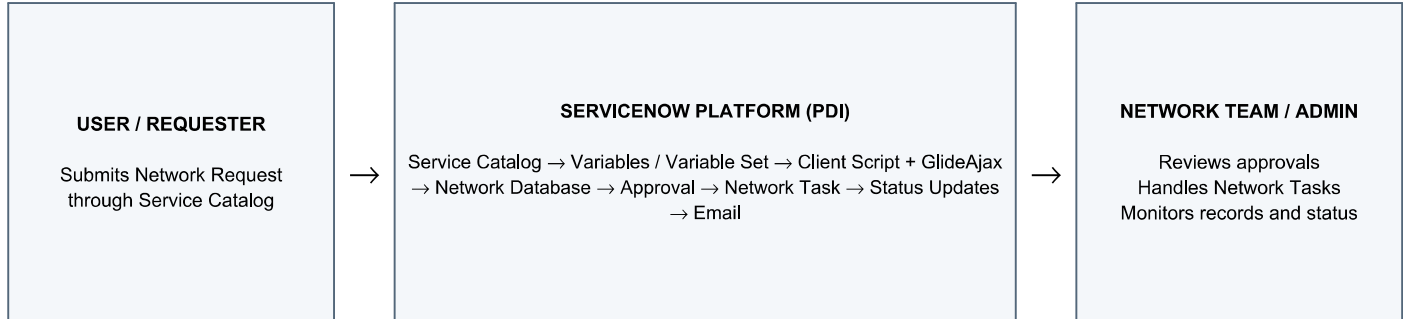


# Project Design Phase-II

## Technology Stack (Architecture & Stack)

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 10-08-2026                                                   |
| Team ID       | 23331A4450                                                   |
| Project Name  | Automated Network Request Management System using ServiceNow |
| Maximum Marks | 4 Marks                                                      |

### Technical Architecture:

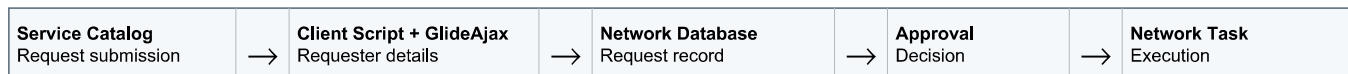


**Table-1 : Components & Technologies:**

| S.No | Component                  | Description                                                               | Technology                                 |
|------|----------------------------|---------------------------------------------------------------------------|--------------------------------------------|
| 1    | User Interface             | Interface used by employees to submit network service requests.           | ServiceNow Service Catalog, Service Portal |
| 2    | Request Data               | Collects requester, device, address, document and request details.        | Catalog Variables, Variable Set            |
| 3    | Client-side Automation     | Automatically populates requester information and controls form behavior. | Catalog Client Script, UI Policy           |
| 4    | Server-side Data Retrieval | Retrieves user details securely from the server.                          | Script Include, GlideAjax                  |
| 5    | Application Database       | Stores submitted network request information and workflow status.         | Custom Network Database Table              |
| 6    | Task Management            | Stores work assigned to the network team after approval.                  | Custom Network Task Table                  |
| 7    | Workflow Automation        | Automates record creation, approvals, updates and notifications.          | Flow Designer                              |
| 8    | Approval Management        | Handles Network Database and Network Task approval decisions.             | ServiceNow Approvals / Flow Designer       |
| 9    | Notifications              | Sends confirmation, approval/completion and rejection emails.             | Send Email / Notification Engine           |
| 10   | Security                   | Controls access to application data and operations.                       | ACLs, Roles                                |
| 11   | Source Control             | Maintains application configuration in version control.                   | ServiceNow Source Control, GitHub          |

**Table-2: Application Characteristics:**

| S.No | Characteristics                  | Description                                                                                           | Technology                                              |
|------|----------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1    | <b>Client-side Customization</b> | Improves form usability by populating user details and applying conditional field behavior.           | JavaScript, Catalog Client Scripts, Catalog UI Policies |
| 2    | <b>Server-side Processing</b>    | Separates server-side user-data retrieval from client-side form logic.                                | Script Includes, GlideAjax                              |
| 3    | <b>Security Implementation</b>   | Provides controlled access to application data and functions.                                         | ServiceNow ACLs, User Roles                             |
| 4    | <b>Workflow Automation</b>       | Coordinates approvals, task creation, record updates and notifications.                               | Flow Designer, Approvals                                |
| 5    | <b>Traceability</b>              | Maintains separate request and task records with approval and work status.                            | Network Database, Network Task tables                   |
| 6    | <b>Maintainability</b>           | Uses modular ServiceNow configuration components that can be updated independently.                   | Scoped Application, Source Control                      |
| 7    | <b>Availability</b>              | Runs on the ServiceNow cloud platform and is accessible through the configured application interface. | ServiceNow Cloud Platform                               |
| 8    | <b>Notifications</b>             | Communicates important workflow outcomes to the requester.                                            | Email / Notification Engine                             |

**Technology Flow:****References:**

ServiceNow Service Catalog documentation – <https://www.servicenow.com/docs/>    ServiceNow Flow Designer documentation – <https://www.servicenow.com/docs/>    ServiceNow Developer documentation – <https://developer.servicenow.com>